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Steering assembly for micromobility transit vehicles

Abstract

In one embodiment, a micromobility transit vehicle may include a steering assembly. The steering assembly may include a steer tube configured to rotate within a headtube of the micromobility transit vehicle, and a stiffening brace positioned about the headtube. A stem may be coupled to the steer tube and the stiffening brace and configured to rotationally lock the stem and the stiffening brace together. A preload assembly may be coupled to the steering assembly and the stem and configured to adjust a preload of the steering assembly.

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Background/Summary

PRIORITY (1) This application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 17/450,116, filed 6 Oct. 2021, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 63/127,659, filed 18 Dec. 2020, which are incorporated herein by reference.

TECHNICAL FIELD

(1) One or more embodiments of the present disclosure relate generally to steering assemblies and more particularly, for example, to a steering assembly for micromobility transit vehicles.

BACKGROUND

(2) Micromobility transit vehicles for hire (e.g., shared scooter, sit-scooters, bicycles, etc.) are subject to repeated use under a wide range of load requirements (e.g., varying rider weights and loads attached to the vehicle, etc.). The steering assembly must be robust to account for the expected loads. To satisfy the load requirements, some steering designs are large in both mass and packaging volume.

(3) Therefore, there is a need in the art for systems and methods that address the deficiencies noted above, other deficiencies known in the industry, or at least offers an alternative to current techniques. For example, improvements are needed for a steering assembly design that densifies one or more mechanical connections to reduce both mass and packaging volume compared to legacy designs.

SUMMARY

(4) Techniques are disclosed for a steering assembly for micromobility transit vehicles. In accordance with one or more embodiments, a micromobility transit vehicle is provided. The micromobility transit vehicle may include a steering assembly. The steering assembly may include a fork, a steer tube extending from the fork, a stiffening brace extending from the fork, and a stem coupled to the steer tube and the stiffening brace. The steer tube may be positioned to rotate within a headtube of the micromobility transit vehicle. The stiffening brace may be positioned about the headtube. The stiffening brace may include a first engagement surface. The stem may include a second engagement surface complementary to the first engagement surface and engaged with the first engagement surface to rotationally lock the stem and stiffening brace together.

(5) One or more embodiments provide a method of coupling a steering assembly to a micromobility transit vehicle. The method may include positioning a steer tube of a steering assembly through a headtube of a frame of a micromobility transit vehicle. The steer tube may extend from a fork of the steering assembly. The method may include positioning a stiffening brace of the steering assembly about the headtube. The stiffening brace may extend from the fork. The method may include coupling a stem to the steer tube and the stiffening brace to secure the steering assembly to the headtube. The coupling may include engaging complementary engagement surfaces of the stem and the stiffening brace to rotationally lock the stem and stiffening brace together.

(6) Additional features are set forth in part in the description that follows and will become apparent to those skilled in the art upon examination of the specification and drawings or may be learned by the practice of the disclosed subject matter. A further understanding of the nature and advantages of the present disclosure may be realized by reference to the remaining portions of the specification and the drawings, which forms a part of this disclosure.

(7) One of skill in the art will understand that each of the various aspects and features of the disclosure may advantageously be used separately in some instances, or in combination with other aspects and features of the disclosure in other instances. In addition, various aspects and features may be omitted. Accordingly, individual aspects can be claimed separately or in combination with other aspects and features. Thus, the present disclosure is merely exemplary in nature and is in no way intended to limit the claimed invention or its applications or uses. It is to be understood that structural and/or logical changes may be made without departing from the spirit and scope of the present disclosure.

(8) The present disclosure is set forth in various levels of detail and no limitation as to the scope of the claimed subject matter is intended by either the inclusion or non-inclusion of elements, components, or the like in this summary. In certain instances, details that are not necessary for an understanding of the disclosure or that render other details difficult to perceive may have been omitted. Moreover, for the purposes of clarity, detailed descriptions of certain features will not be discussed when they would be apparent to those with skill in the art so as not to obscure the description of the present disclosure. The claimed subject matter is not necessarily limited to the arrangements illustrated herein, with the scope of the present disclosure is defined only by the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a block diagram of a portion of a dynamic transportation matching system including a transit vehicle, in accordance with an embodiment of the disclosure.

(2) FIG. 2 illustrates a block diagram of a dynamic transportation matching system incorporating a variety of transportation modalities, in accordance with an embodiment of the disclosure.

- (3) FIGS. 3A, 3B, and 3C illustrate respective diagrams of micromobility transit vehicles for use in a dynamic transportation matching system, in accordance with an embodiment of the disclosure.
- (4) FIG. 3D illustrates a diagram of a docking station for docking one or more micromobility transit vehicles, in accordance with an embodiment of the disclosure.
- (5) FIG. 4 illustrates a diagram of a user interface associated with a micromobility transit vehicle, in accordance with an embodiment of the disclosure.
- (6) FIGS. 5A, 5B, and 5C illustrate respective diagrams of various examples of information rendered on a display of the user interface of FIG. 4, in accordance with an embodiment of the disclosure.
- (7) FIG. 6 illustrates a side view of a micromobility transit vehicle, in accordance with an embodiment of the disclosure.
- (8) FIG. 7 illustrates a front diagram of a steering assembly, in accordance with an embodiment of the disclosure.
- (9) FIG. 8 illustrates a rear diagram of the steering assembly, in accordance with an embodiment of the disclosure.
- (10) FIG. 9 illustrates a front diagram of the steering assembly with a portion of the steering assembly shown transparent for illustration purposes, in accordance with an embodiment of the disclosure.
- (11) FIG. 10 illustrates a rear diagram of the steering assembly with other portions of the steering assembly shown transparent for illustration purposes to highlight a preload assembly, in accordance with an embodiment of the disclosure.
- (12) FIG. 11 illustrates a diagram of the steering assembly with a stem of the steering assembly shown transparent for illustration purposes to highlight the connection of the stem with other portions of the steering assembly, in accordance with an embodiment of the disclosure.
- (13) FIG. 12 is a cross-sectional view of the steering assembly taken along line 12-12 of FIG. 11, in accordance with an embodiment of the disclosure.
- (14) FIG. 13 illustrates a diagram of the steering assembly and showing a cockpit housing and storage basket connected to the steering assembly, in accordance with an embodiment of the disclosure.
- (15) FIG. 14 illustrates a cross-sectional view of the micromobility transit vehicle connected to a docking station, in accordance with an embodiment of the disclosure.
- (16) FIG. 15 illustrates a cable routing diagram of the micromobility transit vehicle, in accordance with an embodiment of the disclosure.
- (17) FIG. 16 illustrates a flow diagram of a process of coupling a steering assembly to a micromobility transit vehicle, in accordance with an embodiment of the disclosure.
- (18) Embodiments of the invention and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures.

DETAILED DESCRIPTION

- (19) The present disclosure is generally directed toward a steering design that densifies the mechanical stem connection to, for example, reduce both mass and packaging volume. The steering assembly includes a fork, a steer tube extending from the fork, and a stiffening brace extending from the fork adjacent to the steer tube. The stiffening brace may extend parallel to the steer tube and may be shaped to extend around the headtube of the vehicle's frame. For example, the stiffening brace may include a concave inner surface that is shaped to fit around a convex outer surface of the headtube.
- (20) The steering assembly may include a stem connected to the steer tube and the stiffening brace. The stem may be connected to the stiffening brace via a planar shear joint. The planar shear joint may be low profile to fit beneath a cosmetic cockpit plastic (e.g., housing). The stiffening brace may be secured to the stem using one or more fasteners, with the fasteners received through

vertically slotted holes of the stiffening brace to allow for vertical adjustment (e.g., headset bearing preload). The planar shear joint rotationally locks (e.g., clocks) the stem and stiffening brace together while allowing the steering assembly to rotate about the steer axis defined by the steer tube. The stem may be connected to the steer tube via a pinch joint. For example, after the shear joint is secured, the circumference of the stem's bore may be pinched around the steer tube to complete the mechanical connection of the upper and lower steering.

(21) The steering assembly may include a preload assembly that adjusts the headset bearing preload. The preload assembly may include a cross pin and a bridge threaded to the cross pin (e.g., via a bolt) and adjustable to draw the stem and fork together to apply axial preload to the headset bearings. The cross pin may be defined in or secured to the steer tube. The bridge may engage the stem, such as seated within a portion of the stem.

(22) A lock triangle may be secured to the stiffening brace to couple the micromobility transit vehicle to a docking station. The lock triangle may pivot about a pin to account for ground surface and docking station variances. The lock triangle may be held in a neutral position via a spring. The spring may include a pair of spring arms received within the lock triangle and a loop nested within a cavity disposed in or adjacent to the stiffening brace. The lengths of the spring arms and loop may be selected to provide repeatable positioning of the lock triangle with respect to the ground (e.g., a ground plane).

(23) FIG. 1 illustrates a block diagram of a portion of a dynamic transportation matching system **100** (e.g., system **100**) including a transit vehicle **110** in accordance with an embodiment of the disclosure. In the embodiment shown in FIG. 1, system **100** includes transit vehicle **110** and optionally a user device **130**. In general, transit vehicle **110** may be a passenger vehicle designed to transport a single person (e.g., a micromobility transit vehicle, a transit bike and scooter vehicle, or the like) or a group of people (e.g., a typical car or truck). More specifically, transit vehicle **110** may be implemented as a motorized or electric kick scooter, bicycle, and/or motor scooter designed to transport one or perhaps two people at once typically on a paved road (collectively, micromobility transit vehicles), as a typical automobile configured to transport up to 4, 7, or 10 people at once, or according to a variety of different transportation modalities (e.g., transportation mechanisms). Transit vehicles similar to transit vehicle **110** may be owned, managed, and/or serviced primarily by a fleet manager/servicer providing transit vehicle **110** for rental and use by the public as one or more types of transportation modalities offered by a dynamic transportation matching system, for example. In some embodiments, transit vehicles similar to transit vehicle **110** may be owned, managed, and/or serviced by a private owner using the dynamic transportation matching system to match their vehicle to a transportation request, such as with ridesharing or ridesourcing applications typically executed on a mobile user device, such as user device **130** as described herein. User device **130** may be a smartphone, tablet, near field communication (NFC) or radio-frequency identification (RFID) enabled smart card, or other personal or portable computing and/or communication device that may be used to facilitate rental and/or operation of transit vehicle **110**.

(24) As shown in FIG. 1, transit vehicle **110** may include one or more of a controller **112**, a user interface **113**, an orientation sensor **114**, a gyroscope/accelerometer **116**, a global navigation satellite system (GNSS) receiver **118**, a wireless communications module **120**, a camera **148**, a propulsion system **122**, an air quality sensor **150**, and other modules **126**. Operation of transit vehicle **110** may be substantially manual, autonomous, and/or partially or completely controlled by user device **130**, which may include one or more of a user interface **132**, a wireless communications module **134**, a camera **138**, and other modules **136**. In other embodiments, transit vehicle **110** may include any one or more of the elements of user device **130**. In some embodiments, one or more of the elements of system **100** may be implemented in a combined housing or structure that can be coupled to or within transit vehicle **110** and/or held or carried by a user of system **100**, such as a transportation requester or rider.

(25) Controller **112** may be implemented as any appropriate logic device (e.g., processing device, microcontroller, processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), memory storage device, memory reader, or other device or combinations of devices) that may be adapted to execute, store, and/or receive appropriate instructions, such as software instructions implementing a control loop for controlling various operations of transit vehicle **110** and/or other elements of system **100**, for example. Such software instructions may also implement methods for processing images and/or other sensor signals or data, determining sensor information, providing user feedback (e.g., through user interface **113** or **132**), querying devices for operational parameters, selecting operational parameters for devices, or performing any of the various operations described herein (e.g., operations performed by logic devices of various devices of system **100**).

(26) In addition, a non-transitory medium may be provided for storing machine readable instructions for loading into and execution by controller **112**. In these and other embodiments, controller **112** may be implemented with other components where appropriate, such as volatile memory, non-volatile memory, one or more interfaces, and/or various analog and/or digital components for interfacing with devices of system **100**. For example, controller **112** may be adapted to store sensor signals, sensor information, parameters for coordinate frame transformations, calibration parameters, sets of calibration points, and/or other operational parameters, over time, for example, and provide such stored data to a transportation requester or rider via user interface **113** or **132**. In some embodiments, controller **112** may be integrated with one or more other elements of transit vehicle **110**, for example, or distributed as multiple logic devices within transit vehicle **110** and/or user device **130**.

(27) In some embodiments, controller **112** may be configured to substantially continuously monitor and/or store the status of and/or sensor data provided by one or more elements of transit vehicle **110** and/or user device **130**, such as the position and/or orientation of transit vehicle **110** and/or user device **130**, for example, and the status of a communication link established between transit vehicle **110** and/or user device **130**. Such communication links may be established and then provide for transmission of data between elements of system **100** substantially continuously throughout operation of system **100**, where such data includes various types of sensor data, control parameters, and/or other data.

(28) User interface **113** of transit vehicle **110** may be implemented as one or more of a display, a touch screen, a keyboard, a mouse, a joystick, a knob, a steering wheel, a yoke, and/or any other device capable of accepting user input and/or providing feedback to a user. In various embodiments, user interface **113** may be adapted to provide user input (e.g., as a type of signal and/or sensor information transmitted by wireless communications module **134** of user device **130**) to other devices of system **100**, such as controller **112**. User interface **113** may also be implemented with one or more logic devices (e.g., similar to controller **112**) that may be adapted to store and/or execute instructions, such as software instructions, implementing any of the various processes and/or methods described herein. For example, user interface **113** may be adapted to form communication links, transmit and/or receive communications (e.g., infrared images and/or other sensor signals, control signals, sensor information, user input, and/or other information), for example, or to perform various other processes and/or methods described herein.

(29) In one embodiment, user interface **113** may be adapted to display a time series of various sensor information and/or other parameters as part of or overlaid on a graph or map, which may be referenced to a position and/or orientation of transit vehicle **110** and/or other elements of system **100**. For example, user interface **113** may be adapted to display a time series of positions, headings, and/or orientations of transit vehicle **110** and/or other elements of system **100** overlaid on a geographical map, which may include one or more graphs indicating a corresponding time series of actuator control signals, sensor information, and/or other sensor and/or control signals. In some embodiments, user interface **113** may be adapted to accept user input including a user-defined

target heading, waypoint, route, and/or orientation, for example, and to generate control signals to cause transit vehicle **110** to move according to the target heading, route, and/or orientation. In other embodiments, user interface **113** may be adapted to accept user input modifying a control loop parameter of controller **112**, for example.

(30) Orientation sensor **114** may be implemented as one or more of a compass, float, accelerometer, and/or other device capable of measuring an orientation of transit vehicle **110** (e.g., magnitude and direction of roll, pitch, and/or yaw, relative to one or more reference orientations such as gravity and/or Magnetic North), camera **148**, and/or other elements of system **100**, and providing such measurements as sensor signals and/or data that may be communicated to various devices of system **100**. Gyroscope/accelerometer **116** may be implemented as one or more electronic sextants, semiconductor devices, integrated chips, accelerometer sensors, accelerometer sensor systems, or other devices capable of measuring angular velocities/accelerations and/or linear accelerations (e.g., direction and magnitude) of transit vehicle **110** and/or other elements of system **100** and providing such measurements as sensor signals and/or data that may be communicated to other devices of system **100** (e.g., user interface **132**, controller **112**).

(31) GNSS receiver **118** may be implemented according to any global navigation satellite system, including a GPS, GLONASS, and/or Galileo based receiver and/or other device capable of determining absolute and/or relative position of transit vehicle **110** (e.g., or an element of transit vehicle **110**) based on wireless signals received from space-born and/or terrestrial sources (e.g., eLoran, and/or other at least partially terrestrial systems), for example, and capable of providing such measurements as sensor signals and/or data (e.g., coordinates) that may be communicated to various devices of system **100**. In some embodiments, GNSS receiver **118** may include an altimeter, for example, or may be used to provide an absolute altitude.

(32) Wireless communications module **120** may be implemented as any wireless communications module configured to transmit and receive analog and/or digital signals between elements of system **100**. For example, wireless communications module **120** may be configured to directly or indirectly receive control signals and/or data from user device **130** and provide them to controller **112** and/or propulsion system **122**. In other embodiments, wireless communications module **120** may be configured to receive images and/or other sensor information (e.g., still images or video images) and relay the sensor data to controller **112** and/or user device **130**. In some embodiments, wireless communications module **120** may be configured to support spread spectrum transmissions, for example, and/or multiple simultaneous communications channels between elements of system **100**. Wireless communication links formed by wireless communications module **120** may include one or more analog and/or digital radio communication links, such as WiFi, Bluetooth, NFC, RFID, and others, as described herein, and may be direct communication links established between elements of system **100**, for example, or may be relayed through one or more wireless relay stations configured to receive and retransmit wireless communications. In various embodiments, wireless communications module **120** may be configured to support wireless mesh networking, as described herein.

(33) In some embodiments, wireless communications module **120** may be configured to be physically coupled to transit vehicle **110** and to monitor the status of a communication link directly or indirectly established between transit vehicle **110** and/or user device **130**. Such status information may be provided to controller **112**, for example, or transmitted to other elements of system **100** for monitoring, storage, or further processing, as described herein. In addition, wireless communications module **120** may be configured to determine a range to another device, such as based on time of flight, and provide such range to the other device and/or controller **112**.

Communication links established by communication module **120** may be configured to transmit data between elements of system **100** substantially continuously throughout operation of system **100**, where such data includes various types of sensor data, control parameters, and/or other data, as described herein.

(34) Propulsion system **122** may be implemented as one or more motor-based propulsion systems, and/or other types of propulsion systems that can be used to provide motive force to transit vehicle **110** and/or to steer transit vehicle **110**. In some embodiments, propulsion system **122** may include elements that can be controlled (e.g., by controller **112** and/or user interface **113**) to provide motion for transit vehicle **110** and to provide an orientation for transit vehicle **110**. In various embodiments, propulsion system **122** may be implemented with a portable power supply, such as a battery. In some embodiments, propulsion system **122** may be implemented with a combustion engine/generator and fuel supply.

(35) For example, in some embodiments, such as when propulsion system **122** is implemented by an electric motor (e.g., as with many micromobility transit vehicles), transit vehicle **110** may include battery **124**. Battery **124** may be implemented by one or more battery cells (e.g., lithium ion battery cells) and be configured to provide electrical power to propulsion system **122** to propel transit vehicle **110**, for example, as well as to various other elements of system **100**, including controller **112**, user interface **113**, and/or wireless communications module **120**. In some embodiments, battery **124** may be implemented with its own safety measures, such as thermal interlocks and a fire-resistant enclosure, for example, and may include one or more logic devices, sensors, and/or a display to monitor and provide visual feedback of a charge status of battery **124** (e.g., a charge percentage, a low charge indicator, etc.).

(36) Other modules **126** may include other and/or additional sensors, actuators, communications modules/nodes, and/or user interface devices, for example, and may be used to provide additional environmental information related to operation of transit vehicle **110**, for example. In some embodiments, other modules **126** may include a humidity sensor, a wind and/or water temperature sensor, a barometer, an altimeter, a radar system, a proximity sensor, a visible spectrum camera or infrared camera (with an additional mount), and/or other environmental sensors providing measurements and/or other sensor signals that can be displayed to a transportation requester or rider and/or used by other devices of system **100** (e.g., controller **112**) to provide operational control of transit vehicle **110** and/or system **100**. In further embodiments, other modules **126** may include a light, such as a head light or indicator light, and/or an audible alarm, both of which may be activated to alert passersby to possible theft, abandonment, and/or other critical statuses of transit vehicle **110**. In particular, and as shown in FIG. 1, other modules **126** may include camera **148** and/or air quality sensor **150**.

(37) Camera **148** may be implemented as an imaging device including an imaging module including an array of detector elements that can be arranged in a focal plane array. In various embodiments, camera **148** may include one or more logic devices (e.g., similar to controller **112**) that can be configured to process imagery captured by detector elements of camera **148** before providing the imagery to communications module **120**. More generally, camera **148** may be configured to perform any of the operations or methods described herein, at least in part, or in combination with controller **112** and/or user interface **113** or **132**.

(38) In various embodiments, air quality sensor **150** may be implemented as an air sampling sensor configured to determine an air quality of an environment about transit vehicle **110** and provide corresponding air quality sensor data. Air quality sensor data provided by air quality sensor **150** may include particulate count, methane content, ozone content, and/or other air quality sensor data associated with common street level sensitivities and/or health monitoring typical when in a street level environment, such as that experienced when riding on a typical micromobility transit vehicle, as described herein.

(39) Transit vehicles implemented as micromobility transit vehicles may include a variety of additional features designed to facilitate fleet management and rider and environmental safety. For example, as shown in FIG. 1, transit vehicle **110** may include one or more of docking mechanism **140**, operator safety measures **142**, vehicle security device **144**, and/or user storage **146**, as described in more detail herein by reference to FIGS. 3A-C.

(40) User interface **132** of user device **130** may be implemented as one or more of a display, a touch screen, a keyboard, a mouse, a joystick, a knob, a steering wheel, a yoke, and/or any other device capable of accepting user input and/or providing feedback to a user, such as a transportation requester or rider. In various embodiments, user interface **132** may be adapted to provide user input (e.g., as a type of signal and/or sensor information transmitted by wireless communications module **134** of user device **130**) to other devices of system **100**, such as controller **112**. User interface **132** may also be implemented with one or more logic devices (e.g., similar to controller **112**) that may be adapted to store and/or execute instructions, such as software instructions, implementing any of the various processes and/or methods described herein. For example, user interface **132** may be adapted to form communication links, transmit and/or receive communications (e.g., infrared images and/or other sensor signals, control signals, sensor information, user input, and/or other information), for example, or to perform various other processes and/or methods described herein.

(41) In one embodiment, user interface **132** may be adapted to display a time series of various sensor information and/or other parameters as part of or overlaid on a graph or map, which may be referenced to a position and/or orientation of transit vehicle **110** and/or other elements of system **100**. For example, user interface **132** may be adapted to display a time series of positions, headings, and/or orientations of transit vehicle **110** and/or other elements of system **100** overlaid on a geographical map, which may include one or more graphs indicating a corresponding time series of actuator control signals, sensor information, and/or other sensor and/or control signals. In some embodiments, user interface **132** may be adapted to accept user input including a user-defined target heading, waypoint, route, and/or orientation, for example, and to generate control signals to cause transit vehicle **110** to move according to the target heading, route, and/or orientation. In other embodiments, user interface **132** may be adapted to accept user input modifying a control loop parameter of controller **112**, for example.

(42) Wireless communications module **134** may be implemented as any wireless communications module configured to transmit and receive analog and/or digital signals between elements of system **100**. For example, wireless communications module **134** may be configured to directly or indirectly transmit control signals from user interface **132** to wireless communications module **120** or **134**. In some embodiments, wireless communications module **134** may be configured to support spread spectrum transmissions, for example, and/or multiple simultaneous communications channels between elements of system **100**. In various embodiments, wireless communications module **134** may be configured to monitor the status of a communication link established between user device **130** and/or transit vehicle **110** (e.g., including packet loss of transmitted and received data between elements of system **100**, such as with digital communication links), and/or determine a range to another device, as described herein. Such status information may be provided to user interface **132**, for example, or transmitted to other elements of system **100** for monitoring, storage, or further processing, as described herein. In various embodiments, wireless communications module **134** may be configured to support wireless mesh networking, as described herein.

(43) Other modules **136** of user device **130** may include other and/or additional sensors, actuators, communications modules/nodes, and/or user interface devices used to provide additional environmental information associated with user device **130**, for example. In some embodiments, other modules **136** may include a humidity sensor, a wind and/or water temperature sensor, a barometer, a radar system, a visible spectrum camera, an infrared camera, a GNSS receiver, and/or other environmental sensors providing measurements and/or other sensor signals that can be displayed to a transportation requester or rider and/or used by other devices of system **100** (e.g., controller **112**) to provide operational control of transit vehicle **110** and/or system **100** or to process sensor data to compensate for environmental conditions. As shown in FIG. 1, other modules **136** may include camera **138**.

(44) Camera **138** may be implemented as an imaging device including an imaging module including an array of detector elements that can be arranged in a focal plane array. In various

embodiments, camera **138** may include one or more logic devices (e.g., similar to controller **112**) that can be configured to process imagery captured by detector elements of camera **138** before providing the imagery to communications module **120**. More generally, camera **138** may be configured to perform any of the operations or methods described herein, at least in part, or in combination with controller **138** and/or user interface **113** or **132**.

(45) In general, each of the elements of system **100** may be implemented with any appropriate logic device (e.g., processing device, microcontroller, processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), memory storage device, memory reader, or other device or combinations of devices) that may be adapted to execute, store, and/or receive appropriate instructions, such as software instructions implementing a method for providing sensor data and/or imagery, for example, or for transmitting and/or receiving communications, such as sensor signals, sensor information, and/or control signals, between one or more devices of system **100**.

(46) In addition, one or more non-transitory mediums may be provided for storing machine readable instructions for loading into and execution by any logic device implemented with one or more of the devices of system **100**. In these and other embodiments, the logic devices may be implemented with other components where appropriate, such as volatile memory, non-volatile memory, and/or one or more interfaces (e.g., inter-integrated circuit (I2C) interfaces, mobile industry processor interfaces (MIPI), joint test action group (JTAG) interfaces (e.g., IEEE 1149.1 standard test access port and boundary-scan architecture), and/or other interfaces, such as an interface for one or more antennas, or an interface for a particular type of sensor).

(47) Sensor signals, control signals, and other signals may be communicated among elements of system **100** and/or elements of other systems similar to system **100** using a variety of wired and/or wireless communication techniques, including voltage signaling, Ethernet, WiFi, Bluetooth, Zigbee, Xbee, Micronet, Near-field Communication (NFC) or other medium and/or short range wired and/or wireless networking protocols and/or implementations, for example. In such embodiments, each element of system **100** may include one or more modules supporting wired, wireless, and/or a combination of wired and wireless communication techniques, including wireless mesh networking techniques. In some embodiments, various elements or portions of elements of system **100** may be integrated with each other, for example, or may be integrated onto a single printed circuit board (PCB) to reduce system complexity, manufacturing costs, power requirements, coordinate frame errors, and/or timing errors between the various sensor measurements.

(48) Each element of system **100** may include one or more batteries, capacitors, or other electrical power storage devices, for example, and may include one or more solar cell modules or other electrical power generating devices. In some embodiments, one or more of the devices may be powered by a power source for transit vehicle **110**, using one or more power leads. Such power leads may also be used to support one or more communication techniques between elements of system **100**.

(49) FIG. 2 illustrates a block diagram of a dynamic transportation matching system **200** (or multimodal transportation system) incorporating a variety of transportation modalities in accordance with an embodiment of the disclosure. For example, as shown in FIG. 2, dynamic transportation matching system **200** may include multiple embodiments of system **100**. In the embodiment shown in FIG. 2, dynamic transportation matching system **200** includes a management system/server **240** in communication with a number of transit vehicles **110a-d** and user devices **130a-b** over a combination of a typical wide area network (WAN) **250**, WAN communication links **252** (solid lines), a variety of mesh network communication links **254** (curved dashed lines), and NFC, RFID, and/or other local communication links **256** (curved solid lines). Dynamic transportation matching system **200** also includes a public transportation status system **242** in communication with a variety of public transportation vehicles, including one or more buses **210a**, trains **210b**, and/or other public transportation modalities, such as ships, ferries, light rail, subways,

streetcars, trolleys, cable cars, monorails, tramways, and aircraft. As shown in FIG. 2, all transit vehicles are able to communicate directly to WAN 250 and, in some embodiments, may be able to communicate across mesh network communication links 254, to convey fleet data and/or fleet status data amongst themselves and/or to and from management system 240.

(50) In FIG. 2, user device 130a may receive an input with a request for transportation with one or more transit vehicles 110a-d and/or public transportation vehicles 210a-b. For example, the transportation request may be a request to use (e.g., hire or rent) one of transit vehicles 110a-d. The transportation request may be transmitted to management system 240 over WAN 250, allowing management system 240 to poll status of transit vehicles 110a-d and to select one of transit vehicles 110a-d to fulfill the transportation request. Upon or after one of the transit vehicles 110a-d is selected to fulfill the transportation request, a fulfillment notice from management system 240 and/or from the selected transit vehicle 110a-d may be transmitted to the user device 130a. In some embodiments, navigation instructions to proceed to or otherwise meet with the selected transit vehicle 110a-d may be sent to the user device 130a. A similar process may occur using user device 130b, but where the transportation request enables a transit vehicle over a local communication link 256, as shown.

(51) Management system 240 may be implemented as a server with controllers, user interfaces, communications modules, and/or other elements similar to those described with respect to system 100 of FIG. 1, but with sufficient processing and storage resources to manage operation of dynamic transportation matching system 200, including monitoring statuses of transit vehicles 110a-d, as described herein. In some embodiments, management system 240 may be implemented in a distributed fashion and include multiple separate server embodiments linked communicatively to each other direction and/or through WAN 250. WAN 250 may include one or more of the Internet, a cellular network, and/or other wired or wireless WANs. WAN communication links 252 may be wired or wireless WAN communication links, and mesh network communication links 254 may be wireless communication links between and among transit vehicles 110a-d, as described herein.

(52) User device 130a in FIG. 2 includes a display of user interface 132 that shows a planned route for a transportation requester or rider attempting to travel from an origination point 260 to a destination 272 using different transportation modalities (e.g., a planned multimodal route), as depicted in a route/street map 286 rendered by user interface 132. For example, management system 240 may be configured to monitor statuses of all available transportation modalities (e.g., including transit vehicles and public transportation vehicles) and provide a planned multimodal route from origination point 260 to destination 272. Such a planned multimodal route may include, for example, a walking route 262 from origination point 260 to a bus stop 264, a bus route 266 from bus stop 264 to a bus stop 268 (e.g., using one or more of transit vehicles 210a or 210b), and a micromobility route 270 (e.g., using one or more of micromobility transit vehicles 110b, 110c, or 110d) from bus stop 268 to destination 272. Also shown rendered by user interface 132 are a present location indicator 280 (indicating a present absolute position of user device 130a on street map 286), a navigation destination selector/indicator 282 (e.g., configured to allow a transportation requester or rider to input a desired navigation destination), and a notice window 284 (e.g., used to render vehicle status data or other information, including user notices and/or alerts, as described herein). For example, a transportation requester or rider may use navigation destination selector/indicator 282 to provide and/or change destination 272, as well as change any portion (e.g., leg, route, etc.) or modality of the multimodal route from origination point 260 to destination 272. In some embodiments, notice window 284 may display instructions for traveling to a next waypoint along the determined multimodal route (e.g., directions to walk to a bus stop, directions to ride a micromobility transit vehicle to a next stop along the route, etc.).

(53) In various embodiments, management system 240 may be configured to provide or suggest an optimal multimodal route to a transportation requester or rider (e.g., initially and/or while traversing a particular planned route), and a transportation requester or rider may select or make

changes to such a route through manipulation of user device **130a**, as shown. For example, management system **240** may be configured to suggest a quickest route, a least expensive route, a most convenient route (to minimize modality changes or physical actions a transportation requester or rider must take along the route), an inclement weather route (e.g., that keeps the transportation requester or rider protected from inclement weather a maximum amount of time during route traversal), or some combination of those that is determined as best suited to the transportation requester or rider, such as based on various user preferences. Such preferences may be based on prior use of system **200**, prior user trips, a desired arrival time and/or departure time (e.g., based on user input or obtained through a user calendar or other data source), or specifically input or set by a user (e.g., a transportation requester or rider) for the specific route, for example, or in general. In one example, origination point **260** may be extremely congested or otherwise hard to access by a ride-share transit vehicle, which could prevent or significantly increase a wait time for the transportation requester or rider and a total trip time to arrive at destination **272**. In such circumstances, a planned multimodal route may include directing the transportation requester or rider to walk and/or take a scooter/bike to an intermediate and less congested location to meet a reserved ride-share vehicle, which would allow the transportation requester or rider to arrive at destination **272** quicker than if the ride-share vehicle was forced to meet the transportation requester or rider at origination point **260**. It will be appreciated that numerous different transportation-relevant conditions may exist or dynamically appear or disappear along a planned route that may make it beneficial to use different modes of transportation to arrive at destination **272** efficiently, including changes in traffic congestion and/or other transportation-relevant conditions that occur mid-route, such as an accident along the planned route. Under such circumstances, management system **240** may be configured to adjust a modality or portion of the planned route dynamically in order to avoid or otherwise compensate for the changed conditions while the route is being traversed.

(54) FIGS. **3A**, **3B**, and **3C** illustrate respective diagrams of micromobility transit vehicles **110b**, **110c**, and **110d**, which may be integrated network systems in accordance with an embodiment of the disclosure. For example, transit vehicle **110b** of FIG. **3A** may correspond to a motorized bicycle integrated with the various elements of system **100** and may be configured to participate in dynamic transportation matching system **200** of FIG. **2**. As shown, transit vehicle **110b** includes controller/user interface/wireless communications module **112/113/120** (e.g., integrated with a rear fender of transit vehicle **110b**), propulsion system **122** configured to provide motive power to at least one of the wheels (e.g., a rear wheel **322**) of transit vehicle **110b**, battery **124** for powering propulsion system **122** and/or other elements of transit vehicle **110b**, docking mechanism **140** (e.g., a spade lock assembly) for docking transit vehicle **110b** at a docking station, user storage **146** implemented as a handlebar basket, and vehicle security device (e.g., an embodiment of vehicle security device **144** of FIG. **1**), which may incorporate one or more of a locking cable **144a**, a pin **144b** coupled to a free end of locking cable **144a**, a pin latch/insertion point **144c**, a frame mount **144d**, and a cable/pin holster **144c**, as shown (collectively, vehicle security device **144**). In some embodiments, controller/user interface/wireless communications module **112/113/120** may alternatively be integrated on and/or within a handlebar enclosure **313**, as shown.

(55) In some embodiments, vehicle security device **144** may be implemented as a wheel lock configured to immobilize rear wheel **322** of transit vehicle **110b**, such as by engaging pin **144b** with spokes of rear wheel **322**. In the embodiment shown in FIG. **3A**, vehicle security device **144** may be implemented as a cable lock configured to engage with a pin latch on a docking station, for example, or to wrap around and/or through a secure pole, fence, or bicycle rack and engage with pin latch **144c**. In various embodiments, vehicle security device **144** may be configured to immobilize transit vehicle **110b** by default, thereby requiring a transportation requester or rider to transmit a request to management system **240** (e.g., via user device **130**) to reserve transit vehicle **110b** before attempting to use transit vehicle **110b**. The request may identify transit vehicle **110b**

based on an identifier (e.g., a QR code, a barcode, a serial number, etc.) presented on transit vehicle **110b** (e.g., such as by user interface **113** on a rear fender of transit vehicle **110b**). Once the request is approved, management system **240** may transmit an unlock signal to transit vehicle **110b** (e.g., via network **250**). Upon receiving the unlock signal, transit vehicle **110b** (e.g., controller **112** of transit vehicle **110b**) may release vehicle security device **144** and unlock rear wheel **322** of transit vehicle **110b**.

(56) Transit vehicle **110c** of FIG. 3B may correspond to a motorized sit-scooter integrated with the various elements of system **100** and may be configured to participate in dynamic transportation matching system **200** of FIG. 2. As shown in FIG. 3B, transit vehicle **110c** includes many of the same elements as those discussed with respect to transit vehicle **110b** of FIG. 3A. For example, transit vehicle **110c** may include user interface **113**, propulsion system **122**, battery **124**, controller/wireless communications module/cockpit enclosure **112/120/312**, user storage **146** (e.g., implemented as a storage recess), and operator safety measures **142a** and **142b**, which may be implemented as various types of head lights, programmable light strips, and/or reflective strips.

(57) Transit vehicle **110d** of FIG. 3C may correspond to a motorized stand or kick scooter integrated with the various elements of system **100** and may be configured to participate in dynamic transportation matching system **200** of FIG. 2. As shown in FIG. 3C, transit vehicle **110d** includes many of the same elements as those discussed with respect to transit vehicle **110b** of FIG. 3A. For example, transit vehicle **110d** may include user interface **113**, propulsion system **122**, battery **124**, controller/wireless communications module/cockpit enclosure **112/120/312**, and operator safety measures **140**, which may be implemented as various types programmable light strips and/or reflective strips, as shown.

(58) FIG. 3D illustrates a docking station **300** for docking transit vehicles (e.g., transit vehicles **110c**, **110e**, and **110g**, etc.) according to one embodiment. As shown, docking station **300** may include multiple bicycle docks, such as docks **302a-c**. In this example, a single transit vehicle (e.g., any one of electric bicycles **304a-d**) may dock in each of the docks **302a-e** of the docking station **300**. Each of the docks **302a-e** may include a lock mechanism for receiving and locking docking mechanism **140** of the electric bicycles **304a-d**. In some embodiments, once a transit vehicle is docked in a bicycle dock, the dock may be electronically coupled to the transit vehicle (e.g., controllers **312a-d** of the transit vehicle) via a link such that the transit vehicle and the dock may communicate with each other via the link.

(59) A transportation requester or rider may use a user device (e.g., user device **130**) to use a micromobility transit vehicle **110b-d** that is docked in one of the bicycle docks **302a-e** by transmitting a request to management system **240**. Once the request is processed, management system **240** may transmit an unlock signal to a micromobility transit vehicle **110b-d** docked in the dock and/or the dock via network **250**. The docking station **300** may automatically unlock the lock mechanism to release the micromobility transit vehicle **110b-d** based on the unlock signal. In some embodiments, each of the docks **302a-e** may also be configured to charge batteries (e.g., batteries **324a-c**) of the electric bicycle **304a-d**, respectively, when the electric bicycle **304a-d** are docked at the docks **302a-c**. In some embodiments, docking station **300** may also be configured to transmit information associated with the docking station **300** (e.g., a number of transit vehicles docked at the docking station **300**, charge statuses of the docked transit vehicles, etc.) to the management system **240**.

(60) FIG. 4 illustrates a diagram of a user interface **400** associated with a micromobility transit vehicle **402** in accordance with an embodiment of the disclosure. The micromobility transit vehicle **402** may be similar to any one of transit vehicles **110b**, **110c**, or **110d**, described above. The user interface **400** may be integrated with the micromobility transit vehicle **402**, such as integrated with at least a portion of a cockpit of the micromobility transit vehicle **402**. In some embodiments, the user interface **400** may form at least a portion of an outer housing of the handlebar of the micromobility transit vehicle **402**. The user interface **400** may be visible to the rider during

operation. For instance, the user interface **400** may generally face rearwardly. The user interface **400** may include a display **410** configured to render information or other data. The display **410** may include many configurations, such as being an electronic ink display, although other configurations are contemplated. In other embodiments, the display **410** may be part of a mobile user computing device, such as a smart phone. As such, content, information, and data discussed herein as being presented on the display **410** can also or alternatively be displayed on the user computing device. (61) The user interface **400** may be similar to the user interface **113** or **132** described above. For example, route guidance information, usage cost, battery charge status, vehicle range, or other information related to the micromobility transit vehicle **402** may be rendered on the display **410**. Information related to the operation of the micromobility transit vehicle **402**, such as time information, map information, navigation information, instructions for operation, operational warnings or notifications, among others, may be rendered on the display **410**. For example, one or more notifications may be rendered on the display **410** instructing or reminding the rider to properly lock and/or park the micromobility transit vehicle **402**. In some embodiments, the user interface **400** may present information similar to that described in U.S. patent application Ser. No. 16/578,995, entitled “MICROMOBILITY ELECTRIC VEHICLE WITH ELECTRONIC DEVICE HOLDER AND INTEGRATED DISPLAY,” which is incorporated herein in its entirety for all purposes.

(62) FIGS. 5A, 5B, and 5C illustrate respective diagrams of various examples of information rendered on the display **410** of the user interface **400** in accordance with an embodiment of the disclosure. The display **410** may render various information and different times, such as during operation of the micromobility transit vehicle **402**, which includes starting, during, or ending a trip or prior to starting use or after ending a ride of the micromobility transit vehicle **402**. For example, as shown in FIG. 5A, the display **410** may render one or more prompts, buttons, or selectable commands (hereinafter “options” **500** for sake of convenience, without intent to limit) for selection. The options **500** may prompt user selection to begin a ride, end a ride, pause a ride, or modify a ride, among others. In some embodiments, the options **500** rendered on the display **410** may allow user selection of one or more navigational commands, such as setting a starting location, setting a destination, starting navigational guidance, ending navigational guidance, modifying an existing navigation route, or the like. In some embodiments, the options **500** rendered on the display **410** may allow a transportation requester or rider to unlock the micromobility transit vehicle **402** from a docking station, pair the micromobility transit vehicle **402** to a docking station, request service or maintenance of the micromobility transit vehicle **402**, report issues with the micromobility transit vehicle **402**, and the like. In some embodiments, the options **500** rendered on the display **410** may allow the rider to turn on a head light assembly, turn off the head light assembly, or otherwise control operation of one or more systems of the micromobility transit vehicle **402**.

(63) Referring to FIG. 5B, the display **410** may render one or more notifications **510** related to operation of the micromobility transit vehicle **402**. For instance, the display **410** may render use agreements, local rules and regulations, liability waivers, operation instructions, operation reminders, and the like for acknowledgment by the rider before, during, or after use. Referring to FIG. 5C, the display **410** may render one or more notifications **520** based on a detected condition of the micromobility transit vehicle **402**. For example, the display **410** may render one or more notifications of a detected use violation (e.g., excessive speed detection, traffic signal violation, etc.), parking violation (e.g., on street, within a landscaped area, within a handicapped zone, etc.), lock violation (e.g., free locking, to an improper sign or structure, failure to lock, etc.), or any combination thereof. In other embodiments, the notifications need not be for a violation, but can be for conveying changes during operation of the micromobility transit vehicle **402**, providing warnings of upcoming hazards or congestion along the ride or trip, providing reminders for use or operation, providing messages at the start and/or end of a ride, including positive messages if the user has complied with all use regulations or guidelines during the trip or user account updates,

such as status, number of rides completed, or total distance traveled on the ride or over multiple rides, and offers or advertisements, such as when the micromobility transit vehicle **402** is detected as being stationary or stopped.

(64) FIG. **6** illustrates a side view of a micromobility transit vehicle **600**, in accordance with an embodiment of the disclosure. The micromobility transit vehicle **600** includes a frame **602**, a propulsion system **604**, and a battery **606**. In embodiments, the micromobility transit vehicle **600** may include a cockpit assembly **610** and a storage basket **612**. The micromobility transit vehicle **600** may be similar to any one of transit vehicles **110b**, **110c**, or **110d**, described above. Thus, any description above of like features is incorporated herein for micromobility transit vehicle **600**.

(65) The frame **602** may include many configurations. For example, the frame **602** may include any number of tubes, brackets, and other components, such as a headtube **620**, a bottom bracket **622**, a downtube **624** extending between the headtube **620** and the bottom bracket **622**. The bottom bracket **622** may rotatably support a portion of the propulsion system **604**, such as a crankset configured to drive a rear wheel **630** through reciprocating motion of a rider's legs. In some embodiments, the frame **602** may include other tubes, such as a seat tube supporting a seat. In some embodiments, the frame **602** may include a top tube, though such is not required, as illustrated in FIG. **6**. Thus, the frame **602** may include any number of frame members coupled together to form a support structure for the various components of the micromobility transit vehicle **600**.

(66) The propulsion system **604** may be any system or device operable to provide a motive force to at least one wheel of the micromobility transit vehicle **600** (e.g., the rear wheel **630**) to propel the micromobility transit vehicle **600** across a surface (e.g., a road surface, a sidewalk, a path, a trail, etc.). The propulsion system **604** may be similar to propulsion system **122** described above, such as including an electric motor configured to drive a rotation of a wheel (e.g., the rear wheel **630**) of micromobility transit vehicle **600**. In some embodiments, at least some portions of the propulsion system **604** may be housed within the frame **602** or other portions (e.g., housings) of the micromobility transit vehicle **600**. As shown, the electric motor may define a portion of the rear wheel **630**. For example, the electric motor may define at least a portion of a hub of the rear wheel **630**.

(67) The battery **606** may be configured to provide electric power to the micromobility transit vehicle **600**. For example, the battery **606** may be configured to provide electric power to the propulsion system **604** (e.g., to the electric motor) to propel the micromobility transit vehicle **600**, for example, as well as to various other modules of the micromobility transit vehicle **600**. The battery **606** may be implemented with one or more battery cells, controllers, and/or safety measures. For example, the battery **606** may include thermal interlocks and one or more logic devices, sensors, and/or a display to monitor and provide visual feedback of a charge status of the battery **606** (e.g., a charge percentage, a low charge indicator, etc.). The battery **606** may be similar to battery **124** described above.

(68) Other modules powered by the battery **606** may include other and/or additional sensors, actuators, communications modules, and/or user interface devices, for example. In some embodiments, the other modules may include one or more proximity sensors, cameras, lights (e.g., a headlight, indicator lights, etc.), alarms, environmental sensors, dynamic sensors, propulsion control systems, and the like. In some embodiments, the battery **606** may include an architecture similar to that described in U.S. patent application Ser. No. 16/728,600, filed Dec. 27, 2019, and entitled "VEHICLE BATTERY INTEGRATION SYSTEMS AND METHODS," or in U.S. patent application Ser. No. 16/836,259, filed Mar. 31, 2020, and entitled "VEHICLE BATTERY SECURING SYSTEMS AND METHODS," which are incorporated herein by reference in their entireties.

(69) In embodiments, battery **606** may be receivable within a battery compartment disposed in the frame **602** (e.g., the downtube **624**). As a result, the battery **606** may be integrated into or placed within the frame **602**, rather than exposed. For example, at least a portion of the battery **606** may be

positioned within the battery compartment to secure the battery **606** to the frame **602**. In such embodiments, the battery compartment may be shaped to receive the battery **606** or at least a portion of the battery **606** therein. Such configurations may provide a secure attachment of the battery **606** to the frame **602**. Additionally, or alternatively, such configurations may provide an attachment that is less prone to vandalism and/or damage or at least limits vandalism and/or damage to the battery **606**. For example, receipt of the battery **606** at least partially within the battery compartment may shield vulnerable portions of the battery **606** positioned within the frame **602**. In embodiments, the battery **606** and/or the securement of the battery **606** within the battery compartment may be similar to that described in U.S. patent application Ser. No. 16/917,102, filed Jun. 30, 2020, and entitled “MICROMOBILITY TRANSIT VEHICLE BATTERY CONNECTION AND LOCK SYSTEMS AND METHODS,” which is incorporated herein by reference in its entirety.

(70) The cockpit assembly **610** may provide a functional, intuitive, and distinctive cockpit or user interface for the rider when riding the micromobility transit vehicle **600**. For example, the cockpit assembly **610** may be implemented with several interfaces, components, or features allowing the rider to see, verify, or control one or more operations of the micromobility transit vehicle **600**. The cockpit assembly **610** may form at least part of an outer housing for the micromobility transit vehicle's handlebars. In some embodiments, the cockpit assembly **610** may include user interface **400** described above. In some embodiments, the cockpit assembly **610** may be similar to the cockpit assembly disclosed in U.S. patent application Ser. No. 16/729,070, filed Dec. 27, 2019, and entitled “MICRO-MOBILITY FLEET VEHICLE COCKPIT ASSEMBLY SYSTEMS AND METHODS,” which is incorporated herein in its entirety for all purposes.

(71) The storage basket **612** may be coupled to frame **602**, such as to the headtube **620**, and may provide one or more functional benefits. For instance, the storage basket **612** may be configured to store a rider's belongings during a ride. In some examples, the storage basket **612** may be configured to provide a locking function. For example, the micromobility transit vehicle **600** may be locked or otherwise secured to a docking station or other stationary object via one or more components of the storage basket **612**, such as a lock cable of the storage basket **612**. In some embodiments, the storage basket **612** may be similar to the storage basket disclosed in U.S. patent application Ser. No. 15/930,195, filed May 12, 2020, and entitled “MICROMOBILITY TRANSIT VEHICLE LOCK-TO MECHANISM SYSTEMS AND METHODS,” which is incorporated herein in its entirety for all purposes.

(72) FIG. 7 illustrates a front diagram of a steering assembly **700** of micromobility transit vehicle **600**, in accordance with an embodiment of the disclosure. The steering assembly **700** is configured to steer micromobility transit vehicle **600**. For example, steering assembly **700** is coupled to a front wheel **702** of micromobility transit vehicle **600** to turn the front wheel **702** under user manipulation of steering assembly **700**. For example, steering assembly **700** may be rotated by the rider of micromobility transit vehicle **600** to turn the front wheel **702** and steer the micromobility transit vehicle **600** as desired. In some embodiments, steering assembly **700** may be designed with increased stiffness and strength to account for repeated use under a wide range of load requirements. For example, the steering assembly **700** may be suited for ridesharing use and requirements, although other configurations are contemplated.

(73) The steering assembly **700** may include a fork **708**, a steer tube **710** extending from the fork **708**, and a stiffening brace **712** extending from the fork **708**. The fork **708** may be coupled to the front wheel **702**, such as defining a pair of front dropouts or fork ends to which the front axle is connected to couple the front wheel **702** to the fork **708**. As shown, the fork **708** includes a pair of rigid blades **716** that are joined at the top by a fork crown **718**, although other configurations are contemplated, including single blade and suspension fork configurations.

(74) The steer tube **710** (e.g., referred to as a steerer tube in some examples) may extend from the fork crown **718** to define a structure about which the steering assembly **700** is rotated to steer the

micromobility transit vehicle **600**. For example, the steer tube **710** may be positioned to rotate within the headtube **620** of frame **602**, such as interfacing with the headtube **620** via a pair of headset bearings **724** (e.g., a headset) mounted in the headtube **620**, as detailed below. Rider manipulation of steering assembly **700** (e.g., via a handlebar) may rotate the steer tube **710** within the headtube **620** about a steer axis to steer micromobility transit vehicle **600** as desired.

(75) The stiffening brace **712** (e.g., referred to as a crown extension, a brace, or a brace/stiffening member in some examples) may extend from the fork crown **718** to increase a stiffness or rigidity of steering assembly **700**. For example, stiffening brace **712** may extend parallel to, and offset from, the steer tube **710** to increase a torsional rigidity of steering assembly **700** to account for expected loads experienced by micromobility transit vehicle **600** (e.g., rider weight, loads positioned within storage basket **612**, vehicle abuse, etc.). As a result, the steer tube **710** and stiffening brace **712** may define a double beam structure for increased strength. In embodiments, stiffening brace **712** extend from the fork **708** to be positioned at least partially about the headtube **620**, as detailed below. For example, stiffening brace **712** may be positioned such that rotation of steering assembly **700** about the steer tube **710** rotates the stiffening brace **712** about the headtube **620**.

(76) As shown, the stiffening brace **712** includes a main body **730** extending from the fork **708**, and a terminal end **732** extending from (or defined at the end of) the main body **730**. In embodiments, the terminal end **732** may have a reduced dimension (e.g., thickness) compared to the main body **730** to facilitate attachment of the stiffening brace **712** to the stem, as described below. Depending on the application, the terminal end **732** may be defined as one or multiple tabs (e.g., a pair of tabs), although other configurations are contemplated.

(77) With continued reference to FIG. 7, steering assembly **700** may include a stem **740** coupled to the steer tube **710** and the stiffening brace **712** to secure the steering assembly **700** to the headtube **620**. For example, stem **740** may include a threadless design, with the stem **740** clamped around the steer tube **710**. The stiffening brace **712** may be secured to the stem **740** to complete the double beam structure. As a result, the ends of the stiffening brace **712** and steer tube **710** may be connected to the same structure for increased strength and rigidity.

(78) In embodiments, the connection between the stem **740** and the stiffening brace **712** may be designed to rotationally lock the stem **740** and stiffening brace **712** together. For example, the engagement of the stiffening brace **712** with the stem **740** may clock the stem **740** and stiffening brace **712** together, such as maintaining engagement of the stem **740** with the stiffening brace **712** during rotation of the steering assembly **700**. Depending on the application, the stiffening brace **712** may include a first engagement surface **750**, and the stem **740** may include a second engagement surface **752** complementary to the first engagement surface **750** and engaged with the first engagement surface **750** to rotationally lock the stem **740** and stiffening brace **712** together. As a result, movement of the stem **740** relative to the stiffening brace **712** may be limited or prevented to tie rotation of the stem **740** with rotation of the stiffening brace **712** and/or maintain alignment of the stem **740** with the stiffening brace **712**.

(79) FIG. 8 illustrates a rear diagram of the steering assembly **700**, in accordance with an embodiment of the disclosure. Referring to FIG. 8, the stem **740** may be coupled to the steer tube **710** via a pinch joint **800**. For example, stem **740** may include a split configuration, with one or fasteners **802** clamping the stem **740** onto the steer tube **710**. In such embodiments, the circumference of the stem bore is pinched around the steer tube **710**. In embodiments, stiffening brace **712** may include an inner surface **806** shaped to fit the stiffening brace **712** around the headtube **620**. For example, the inner surface **806** of stiffening brace **712** may be concave and shaped to fit around a convex outer surface of the headtube **620**. The above configurations may allow the stiffening brace **712** to rotate about the headtube **620** as the steering assembly **700** is rotated.

(80) FIG. 9 illustrates a front diagram of the steering assembly **700** with the stem **740** shown

transparent for illustration purposes, in accordance with an embodiment of the disclosure. Referring to FIG. 9, the steering assembly **700** may include a slotted hole **900** disposed in the stiffening brace **712**, such as disposed in the terminal end **732**. For example, a slotted hole **900** may be disposed in each tab of the terminal end **732**. A fastener **902** may be disposed in each slotted hole **900** and coupled to the stem **740** to secure the stiffening brace **712** to the stem **740**. As described in more detail below, the slotted hole(s) **900** may be shaped to allow for axial adjustment of the stem **740** relative to the stiffening brace **712**. For example, the fastener(s) **902** may slide within the slotted hole(s) **900** to allow axial movement of the stem **740** along the steer tube **710** to, for instance, fit the steering assembly **700** to the headtube **620** and/or adjust a bearing preload. Accordingly, the slotted holes **900** may be slotted vertically to allow for vertical adjustment of the stem **740**.

(81) FIG. 10 illustrates a rear diagram of the steering assembly **700** with portions of the steering assembly **700** shown transparent for illustration purposes to highlight a preload assembly, in accordance with an embodiment of the disclosure. Referring to FIG. 10, the steering assembly **700** may include a preload assembly **1000** configured to adjust a headset bearing preload. The preload assembly **1000** may include one or more features to draw the headset bearings **724** together at a desired headset bearing preload. For example, the preload assembly **1000** may include a cross pin **1002** associated with the steer tube **710**, a bridge **1004** engaging the stem **740**, and a bolt **1006** extending between the bridge **1004** and the cross pin **1002**, although other configurations are contemplated.

(82) The cross pin **1002** may be coupled to or defined in the steer tube **710**, such as positioned within the interior bore of the steer tube **710**. The bridge **1004** may engage the top of the stem **740**, such as positioned at least partially within a seat defined in the stem **740**. The bolt **1006** may extend from the bridge **1004** for threaded engagement with cross pin **1002**. The bolt **1006** may be adjusted to adjust the headset bearing preload. For example, a tension of the bolt **1006** may be adjustable to draw the stem **740** and fork **708** together to apply axial preload to the headset bearings **724**. For instance, adjusting the bolt **1006** may draw the steer tube **710** towards the stem **740** to adjust a preload compressing the headset bearings **724** together and into the headtube **620**.

(83) FIG. 11 illustrates a diagram of the steering assembly **700** with the stem **740** of the steering assembly **700** shown transparent for illustration purposes to highlight the connection of the stem **740** with the steer tube **710** and stiffening brace **712**, in accordance with an embodiment of the disclosure. FIG. 12 is a cross-sectional view of the steering assembly **700** taken along line 12-12 of FIG. 11, in accordance with an embodiment of the disclosure. Referring to FIGS. 11-12, each of the first engagement surface **750** and the second engagement surface **752** may be a planar surface (see FIG. 12). As a result, the engagement surfaces may define a planar shear joint **1100** between the stiffening brace **712** and the stem **740** to rotationally lock the stem **740** and stiffening brace **712** together.

(84) As shown, the stem **740** may be connected to the steer tube **710** and stiffening brace **712** via multiple connections. For example, the stem **740** may be connected to the steer tube **710** via a first connection (e.g., the pinch joint **800**), and the stem **740** may be connected to the stiffening brace **712** via a second connection (e.g., the planar shear joint **1100**). To connect the stem **740** to the steering assembly **700**, the stem **740** may be slid at least partially onto the steer tube **710**, with the first engagement surface **750** of stiffening brace **712** aligned and engaged with the second engagement surface **752** of stem **740**. The fasteners **902** may be inserted through the slotted holes **900** of stiffening brace **712** to engage stem **740**. Bridge **1004** may be seated in or onto stem **740**, and bolt **1006** may be threaded to cross pin **1002**. The bolt **1006** may then be adjusted to adjust the headset bearing preload as desired (e.g., to a specification). The fasteners **902** may then be fully tightened to secure stiffening brace **712** to stem **740**, and fasteners **802** may be tightened to fully tighten the stem **740** onto the steer tube **710**.

(85) FIG. 13 illustrates a diagram of the steering assembly **700** and showing a cockpit housing **1300** and storage basket **612** connected to the steering assembly **700**, in accordance with an

embodiment of the disclosure. Referring to FIG. 13, a cockpit housing **1300** of cockpit assembly **610** may be coupled to the steering assembly **700**. For example, the cockpit housing **1300** may be coupled to the stem **740** to at least partially conceal the engagement of the stem **740** with the steer tube **710** and/or the engagement of the stem **740** with the stiffening brace **712**. As shown, the cockpit housing **1300** may be shaped to sit flush with the stiffening brace **712**. For example, the cockpit housing **1300** may sit flush with the main body **730** of stiffening brace **712** to streamline the micromobility transit vehicle **600** and provide a desired aesthetic and/or functional characteristic.

(86) The terminal end **732** of stiffening brace **712** may be shaped to fit beneath the cockpit housing **1300** to, for example, reduce the form factor of cockpit housing **1300**. For instance, the planar shear joint **1100** may be low profile to fit beneath the cockpit housing **1300**. As a result, the mechanical connection between the stiffening brace **712** and the stem **740** may be densified to reduce both mass and packaging volume compared to other designs.

(87) With continued reference to FIG. 13, storage basket **612** may be connected to stiffening brace **712**. For example, as shown in at least FIGS. 8, 9, and 13, stiffening brace **712** may include a basket connection **1308**. The storage basket **612** may be coupled to the basket connection **1308** to secure the storage basket **612** to the micromobility transit vehicle **600**. In this manner, the storage basket **612** may rotate with rotation of steering assembly **700**.

(88) FIG. 14 illustrates a cross-sectional view of the micromobility transit vehicle **600** connected to a docking station **1400**, in accordance with an embodiment of the disclosure. Referring to FIG. 14, micromobility transit vehicle **600** may be configured for engagement with a docking station **1400** to lock and/or charge micromobility transit vehicle **600** when docked. For example, micromobility transit vehicle **600** may include a lock triangle **1402** (or other structure) secured to the stiffening brace **712** and configured to engage docking station **1400**. For instance, lock triangle **1402** may include a hook **1404** that engages a pin and/or plate within docking station **1400** to lock micromobility transit vehicle **600** to docking station **1400**. In some embodiments, engagement of lock triangle **1402** with docking station **1400** may electrically couple micromobility transit vehicle **600** to docking station **1400** to, for example, charge the battery **606** of micromobility transit vehicle **600**. In embodiments, the lock triangle **1402** may be secured to docking station **1400** in a manner similar to that described in U.S. patent application Ser. No. 16/179,564, filed Nov. 2, 2018, and entitled "LOCKING MECHANISM FOR A BICYCLE SHARING SYSTEM," or in U.S. patent application Ser. No. 13/591,627, filed Aug. 22, 2012, and entitled "METHOD AND APPARATUS FOR SECURING A MOVABLE ITEM TO A STRUCTURE," which are hereby incorporated by reference in their entireties.

(89) Referring to FIG. 14, the lock triangle **1402** may be configured to facilitate engagement of lock triangle **1402** with docking station **1400** under various conditions. For example, the lock triangle **1402** may be configured to pivot about a pin **1408** to account for ground surface and docking station variances. In embodiments, lock triangle **1402** may include a spring **1410** configured to bias the lock triangle **1402** to a neutral position about the pin **1408**. For instance, spring **1410** may be sized and/or shaped to support the weight of lock triangle **1402** such that lock triangle **1402** may pivot about pin **1408** up or down to engage docking station **1400**. As shown, the spring **1410** may include a first spring arm **1416** received within the lock triangle **1402** and a second spring arm **1418** nested within a cavity **1424** disposed in or adjacent to the stiffening brace **712**. The lengths of the first and second spring arms **1416**, **1418** may be selected to provide repeatable positioning of the lock triangle **1402** with respect to the ground (e.g., a ground plane). In some embodiments, the second spring arm **1418** may be a loop. In some embodiments, the first spring arm **1416** may be a pair of spring arms defined by ends of the spring **1410**.

(90) FIG. 15 illustrates a cable routing diagram of the micromobility transit vehicle **600**, in accordance with an embodiment of the disclosure. Referring to FIG. 15, one or more cables **1500** may be routed through at least one of the steer tube **710** or the stiffening brace **712** of the steering

assembly **700**. For example, multiple control cables (e.g., electrical, hydraulic, or steel (e.g., stainless steel)) may be routed through the steering assembly **700**. As shown, a headlight cable **1500A** may be routed at least partially through the stiffening brace **712** for connection to a headlight disposed in the cockpit assembly **610** and/or the storage basket **612**. A front brake cable **1500B** may also be routed at least partially through the stiffening brace **712** for connection with a front brake. A rear brake cable **1500C** may be routed at least partially through the steer tube **710** for connection with a rear brake. In some embodiments, the rear brake cable **1500C** may be routed through the downtube **624** to the rear brake. A VCU cable **1500D** may be routed at least partially through the steer tube **710**, such as to route the VCU cable **1500D** from the cockpit assembly **610** to a motor controller or other component of the micromobility transit vehicle **600**. In embodiments, a charge cable **1500E** may be routed through at least a portion of the steering assembly **700** from the lock triangle **1402** to the battery **606**. The above cables **1500** are exemplary only, and other cables may be routed at least partially through the steering assembly **700**.

(91) Referring to FIGS. **7**, **9**, **10**, and **12**, one or more cutouts may be disposed in the steer tube **710** and/or stiffening brace **712** to facilitate routing of the cables **1500** at least partially through the steering assembly **700**. For example, referring to FIGS. **7**, **9**, and **12**, a first cutout **1520** may be disposed in the stiffening brace **712** to facilitate routing of at least one of the headlight cable **1500A**, the charge cable **1500E**, or the front brake cable **1500B** through the stiffening brace **712**. Referring to FIG. **10**, a second cutout **1522** may be disposed in the steer tube **710** to facilitate routing of at least one of the VCU cable **1500D**, the charge cable **1500E**, or the rear brake cable **1500C** through the steer tube **710**.

(92) FIG. **16** illustrates a flow diagram of a process **1600** of coupling a steering assembly to a micromobility transit vehicle, in accordance with an embodiment of the disclosure. It should be appreciated that any step, sub-step, sub-process, or block of process **1600** may be performed in an order or arrangement different from the embodiments illustrated by FIG. **16**. For example, one or more blocks may be omitted from or added to the process **1600**. Although process **1600** is described with reference to the embodiments of FIGS. **1-15**, process **1600** may be applied to other embodiments.

(93) The micromobility transit vehicle and steering assembly associated with process **1600** may be similar to micromobility transit vehicle **600** and steering assembly **700** described above. For example, steering assembly may include a fork, a steer tube extending from the fork, a stiffening brace extending from the fork, and a stem configured to connect to both the steer tube and the stiffening brace. In addition, micromobility transit vehicle may include a frame having a headtube.

(94) In block **1602**, process **1600** may include positioning the steer tube through the headtube of micromobility transit vehicle. For example, the steer tube may be positioned to rotate within the headtube. The steer tube may be rotatably coupled to the steer tube, such as via a pair of headset bearings (e.g., a headset). In this manner, rider manipulation of the steering assembly (e.g., via a handlebar) may rotate the steer tube within the headtube to steer the micromobility transit vehicle as desired.

(95) In block **1604**, process **1600** may include positioning the stiffening brace about the headtube. For example, the stiffening brace may extend parallel to the steer tube and positioned to rotate around the headtube with rotation of steering assembly. The stiffening brace may be low profile and shaped to accommodate rotation about the headtube. For instance, the stiffening brace may include a concave inner surface shaped to fit around a convex outer surface of the headtube.

(96) In block **1606**, process **1600** may include coupling the stem to the steer tube and the stiffening brace to secure the steering assembly to the headtube. Block **1606** may include securing the stem to the steer tube via a pinch joint. For example, the stem may be pinched around the steer tube. Block **1606** may include securing the stem to the stiffening brace via one or more fasteners received within slotted holes disposed in the stiffening brace. The slotted holes may allow for axial adjustment of the stem relative to the stiffening brace, such as allowing axial movement of the stem

along the steer tube.

(97) In block **1608**, process **1600** may include engaging complementary engagement surfaces of the stem and the stiffening brace to rotationally lock the stem and stiffening brace together, such as explained above. For instance, the engagement surfaces of the stem and stiffening brace may be planar to define a planar shear joint between the stem and the stiffening brace to clock (or rotationally lock) the stem and stiffening brace together. Block **1608** may include securing the stem to the stiffening brace via the one or more fasteners to draw the complementary engagement surfaces together.

(98) In block **1610**, process **1600** may include axially adjusting the stem relative to the stiffening brace and along the steer tube. For example, the stem may be slid along the steer tube to seat the stem against the headtube to secure the steering assembly to the frame. The stem may be axially adjusted to align the connection of the stem with the stiffening brace and/or the steer tube. The stem may be axially adjusted to accommodate different headtube dimensions.

(99) In block **1612**, process **1600** may include adjusting a headset bearing preload via a preload assembly. The preload assembly may be similar to preload assembly discussed above. For example, preload assembly may include a cross pin associated with (e.g., disposed in, defined in, coupled to, etc.) the steer tube, a bridge engaging the stem, and a bolt extending between the bridge and the cross pin. Block **1612** may include adjusting a tension of the bolt to draw the stem and fork together to apply axial preload to a pair of headset bearings, such as explained above.

(100) In block **1614**, process **1600** may include routing one or more cables through at least one of the steer tube or the stiffening brace of the steering assembly. For example, one or more electrical or control cables may be routed through the steer tube and/or the stiffening brace between the vehicle's cockpit assembly/housing and various components of the vehicle. Depending on the application, power, data, control, or brake cables may be routed through the steer tube and/or the stiffening brace, such as described above.

(101) In block **1616**, process **1600** may include coupling a cockpit housing to the stem to conceal at least one of the engagement of the stem with the steer tube or the engagement of the stem with the stiffening brace. For example, the cockpit housing may be fastened to the stem to cover the engagements of the stiffening brace and the steer tube with the stem. The cockpit housing may be shaped to sit flush with at least a portion of the stiffening brace to streamline the cockpit design and provide a desired aesthetic and/or functional characteristic. In such embodiments, a terminal end of the stiffening brace may be shaped to fit beneath the cockpit housing to, for instance, reduce the form factor of the cockpit housing.

(102) In block **1618**, process **1600** may include engaging a lock triangle of the micromobility transit vehicle to a docking station. The lock triangle may be similar to lock triangle **1402** described above. For instance, the lock triangle may be rotatable about a pin and include a spring biasing the lock triangle to a neutral position about the pin, as described above. The lock triangle may engage the docking station to secure the micromobility transit vehicle. For instance, the lock triangle may be locked within the docking station to prevent unauthorized use of micromobility transit vehicle. In embodiments, engagement of the lock triangle with docking station may complete an electrical circuit to charge the micromobility transit vehicle (e.g., a battery of micromobility transit vehicle) while the vehicle is docked.

(103) Where applicable, various embodiments provided by the present disclosure can be implemented using hardware, software, or combinations of hardware and software. Also, where applicable, the various hardware components and/or software components set forth herein can be combined into composite components comprising software, hardware, and/or both without departing from the spirit of the present disclosure. Where applicable, the various hardware components and/or software components set forth herein can be separated into sub-components comprising software, hardware, or both without departing from the spirit of the present disclosure. In addition, where applicable, it is contemplated that software components can be implemented as

hardware components, and vice-versa.

(104) Software in accordance with the present disclosure, such as non-transitory instructions, program code, and/or data, can be stored on one or more non-transitory machine-readable mediums. It is also contemplated that software identified herein can be implemented using one or more general purpose or specific purpose computers and/or computer systems, networked and/or otherwise. Where applicable, the ordering of various steps described herein can be changed, combined into composite steps, and/or separated into sub-steps to provide features described herein.

(105) One or more features or functions described herein may be included or omitted, or enabled or disabled on an opt-in basis, particularly for optimizing rider safety, comfort, and reliability. For example, a rider may choose to turn on or turn off, or use or not use, one or more of the features or functions described herein for their personal safety and comfort. In some instances, these features or functions may be turned on temporarily and/or turned on automatically to maximize rider safety and comfort, among various other possibilities.

(106) All relative and directional references (including upper, lower, front, rear, and so forth) are given by way of example to aid the reader's understanding of the examples described herein. They should not be read to be requirements or limitations, particularly as to the position, orientation, or use unless specifically set forth in the claims. Connection references (e.g., attached, coupled, connected, joined, and the like) are to be construed broadly and may include intermediate members between a connection of elements and relative movement between elements. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to each other, unless specifically set forth in the claims.

(107) The present disclosure teaches by way of example and not by limitation. Therefore, the matter contained in the above description or shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. It should also be understood that numerous modifications and variations are possible in accordance with the principles of the present disclosure. The following claims are intended to cover all generic and specific features described herein, as well as all statements of the scope of the present method and system, which, as a matter of language, might be said to fall there between.

Claims

1. A micromobility transit vehicle comprising: a steering assembly comprising: a steer tube configured to rotate within a headtube of the micromobility transit vehicle, and a stiffening brace positioned about the headtube; a stem coupled to the steer tube and the stiffening brace and configured to rotationally lock the stem and the stiffening brace together, wherein the stem is coupled to the stiffening brace via a slotted hole of the stiffening brace, the slotted hole being configured with a shape to allow for axial adjustment of the stem along the steer tube and relative to the stiffening brace; and a preload assembly coupled to the steering assembly and the stem and configured to adjust a preload of the steering assembly.
2. The micromobility transit vehicle of claim 1, wherein the preload assembly is configured to adjust the preload in an axial direction along the steer tube.
3. The micromobility transit vehicle of claim 1, wherein the preload assembly is configured to adjust the preload by moving the steer tube relative to the stem.
4. The micromobility transit vehicle of claim 1, wherein the preload assembly comprises: a cross pin associated with the steer tube; a bridge associated with the stem; and a bolt extending between the bridge and the cross pin, wherein the bolt is adjustable to move the steer tube relative to the stem.
5. The micromobility transit vehicle of claim 4, wherein the cross pin is configured in a bore of the steer tube.

6. The micromobility transit vehicle of claim 4, wherein the bridge is engaged with a top surface of the stem.
 7. The micromobility transit vehicle of claim 4, wherein the stem comprises a seat, and wherein the bridge is configured at least partially within the seat.
 8. The micromobility transit vehicle of claim 4, wherein the bolt is threadedly coupled to the cross pin.
 9. The micromobility transit vehicle of claim 1, further comprising a pair of headset bearings configured in the headtube.
 10. The micromobility transit vehicle of claim 9, wherein the preload assembly is further configured to adjust a preload of the pair of head headset bearings.
 11. The micromobility transit vehicle of claim 1, wherein the stem is coupled to the steer tube via a pinch joint.
 12. The micromobility transit vehicle of claim 1, wherein the stiffening brace comprises a first engagement surface, and wherein the stem comprises a second engagement surface complementary to the first engagement surface and engaged with the first engagement surface.
 13. A method of manufacturing a micromobility transit vehicle, the method comprising: positioning a steer tube of a steering assembly through a headtube of the micromobility transit vehicle; positioning a stiffening brace of the steering assembly about the headtube; coupling a stem to the steer tube and the stiffening brace to secure the steering assembly to the headtube and to rotationally lock the stem and the stiffening brace together, wherein the stem is coupled to the stiffening brace via a slotted hole of the stiffening brace, the slotted hole being configured with a shape to allow for axial adjustment of the stem along the steer tube and relative to the stiffening brace; and coupling a preload assembly to the steering assembly and the stem, wherein the preload assembly is configured to adjust a preload of the steering assembly.
 14. The method of claim 13, further comprising adjusting the preload of the steering assembly in an axial direction along the steer tube via the preload assembly.
 15. The method of claim 13, further comprising adjusting the preload of the steering assembly by moving the steer tube relative to the stem via the preload assembly.
 16. The method of claim 13, wherein the preload assembly comprises: a cross pin associated with the steer tube; a bridge associated with the stem; and a bolt extending between the bridge and the cross pin, wherein the bolt is adjustable to move the steer tube relative to the stem.
 17. The method of claim 16, wherein coupling the preload assembly to the steering assembly and the stem comprises coupling the cross pin to the steer tube and coupling the bridge to the stem.
 18. The method of claim 16, further comprising adjusting the preload of the steering assembly by adjusting the bolt to move the steer tube relative to the stem.
 19. The method of claim 13, further comprising configuring a pair of headset bearings in the headtube, wherein the preload assembly is further configured to adjust a preload of the pair of headset bearings.
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